

Mohammed El Barhichi

AI Engineer | Agentic AI, GenAI Systems & Cloud Infrastructure

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SUMMARY

AI Engineer focused on building production-grade Agentic AI systems and scalable GenAI infrastructure. Experienced in multi-agent orchestration, interoperability protocols, cloud-native AI deployment, and enterprise AI solutions across different industry environments.

PROFESSIONAL EXPERIENCE

AI Engineer - Agentic AI Lead

01/2026 – Present | Paris

- Designed, built and deployed a production **multi-agent** system on Google Agent Engine using ADK for network topology queries and simulation API calls, integrating Spanner Graph, OpenAPI toolsets, and automated evaluation framework to benchmark performance across deployments.
- Led the **A2A-T** initiative, a joint innovation collaboration between Orange and Huawei defining an **agent-to-agent** protocol for telecom industry, enabling autonomous multi-vendor agents to interoperate across network operations, from protocol design to reference implementation.
- Drive **agentic AI** standards and **technical recommendations** on architecture and tooling within the **Innovation Data & AI team**, and contribute to Orange's **Corporate Expertise Center** on GenAI/Agentic AI through technical deep-dives on observability and agent orchestration.
- Stack:** Google ADK, Agent Engine, LangChain, LangSmith, FastAPI, React, MCP, A2A Protocol, GCP (Cloud Run, Spanner Graph), Docker.

AI Software Engineer Intern

05/2025 – 11/2025 | Paris

- Built and deployed a production-grade, voice-enabled **AI assistant** for in-car services, using LangGraph, LLM APIs (OpenAI, Mistral, Llama), and multimodal pipelines (ASR/TTS, APIs, retrieval systems) on GCP and Azure.
- Designed agentic workflows with tool-calling, conversational memory, and context-aware orchestration for low-latency in-vehicle interactions.
- Developed an evaluation and observability framework (**agents-as-judges**) to monitor robustness, latency, hallucination rates, and user impact.
- Collaborated with product and engineering stakeholders to transform prototype AI systems into scalable production services.
- Stack:** FastAPI, LangChain, LangGraph, HuggingFace, Vector DBs (FAISS/Chroma), OpenAI/Mistral APIs, Docker, K8s, CI/CD, Azure, GCP.

Machine Learning Project Lead

10/2024 – 04/2025 | Paris

- Built a real-time low-latency streaming pipeline for real-time detection of adversarial prompts and jailbreak attempts in LLM systems.
- Implemented dynamic model **hot-swapping** and monitoring pipelines to ensure uninterrupted deployment and continuous model evaluation.
- Focused on production AI reliability, latency monitoring, and scalable inference for enterprise conversational systems.
- Stack:** Apache Pulsar, Beam, Spark, Kafka, MLflow, Docker, K8s, Airflow, Terraform, Grafana, GitHub Actions, GCP

GenAI Engineer Intern

04/2024 – 08/2024 | Casablanca

- Designed and delivered end-to-end **RAG-based chatbots** for clients using LangChain, Azure OpenAI, FAISS/Chroma, and LangSmith
- Implemented **advanced retrieval strategies**: semantic chunking, multi-query retrievers, history-aware retrievers, query decomposition, achieving 87–89% answer relevancy in client evaluations
- Built an **observability** and traceability UI and applied RAGAS **evaluation harness** and metrics to ensure reliability and reduce hallucinations.
- Stack:** Python, LangChain/LangGraph, FAISS/Chroma, PostgreSQL, Streamlit, LangSmith, RAGAS Airflow, Azure OpenAI, GCP.

Data Scientist Intern

06/2023 – 09/2023 | Casablanca

- Automated compliance data extraction by developing an **NLP pipeline** combining **web scraping** with **Named Entity Recognition** models
- Achieved **95% reduction** in manual processing time; introduced confidence scoring, error monitoring and retry/fallback strategies.
- Stack:** Python, NLTK/spaCy, BeautifulSoup, Selenium, PostgreSQL/BigQuery, Dash/Gradio.

EDUCATION

Master of Engineering in Artificial Intelligence (Diplôme d'Ingénieur)

2022 – 2025

CentraleSupélec - CentraleCasablanca

Key Courses: Machine Learning, Deep Learning, Reinforcement Learning, Multi-Agent Systems, Computer Vision, Natural Language Processing.

Grade: GPA 4.0/4.0 | **Valedictorian** (Major de Promo)

Exchange Program - Technology, Data & Strategy

01/2024 – 05/2024

ESSEC Business School

Key Courses: Data-driven Decision-Making, Corporate Finance, Business Analytics, Innovation Strategy, Market Analysis, Entrepreneurship.

SKILLS AND LANGUAGES

GenAI / Agentic AI: LangChain, LangGraph, Google ADK, LlamaIndex, MCP, A2A Protocol, Multi-Agent Systems, RAG, RAGAS, Hugging Face Transformers, Vector Databases (FAISS, ChromaDB, Neo4j), Tool Calling, LLM Evaluation, Prompt Engineering, Claude APIs
Cloud, Infrastructure & MLOps: GCP (Agent Engine, Cloud Run, Spanner Graph), AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, MLflow, Grafana, OpenTelemetry, FastAPI, REST APIs
Programming: Python, SQL, GQL, TypeScript, JavaScript, React, Java, Bash, Claude Code, OpenAI Codex
Languages: English (C1 TOEIC / Linguaskill), French (DALF C1, bilingual), Arabic (native).

RELEVANT PROJECTS

Anomaly Detection & Heart Crisis Prediction – Real-Time Health Monitoring Project

- Built a real-time AWS pipeline ingesting smartwatch data via **Kinesis**, preprocessing with AWS **Lambda** and feature extraction in Apache **Spark**.
- Deployed heart-crisis prediction models in **SageMaker**, achieving over 96% specificity and sensitivity; included automated retraining on drift.
- Stored patient data in DynamoDB and captured latency, throughput, and drift metrics in CloudWatch, visualized via dashboards and Grafana.

Facial Emotion Recognition Project - MLOps Project

- Developed and deployed a facial emotion recognition application designed to analyze and detect emotions from facial expressions in real-time.
- Leveraged **CI/CD** technologies to ensure scalability, monitoring, and seamless integration for production use